

Data Flow Mapping

Data Flow Mapping describes how data moves through the system from the user to the database and back. The Minutes of Meeting (MOM) Management System follows a structured data flow to ensure secure, accurate, and efficient processing of meeting-related information.

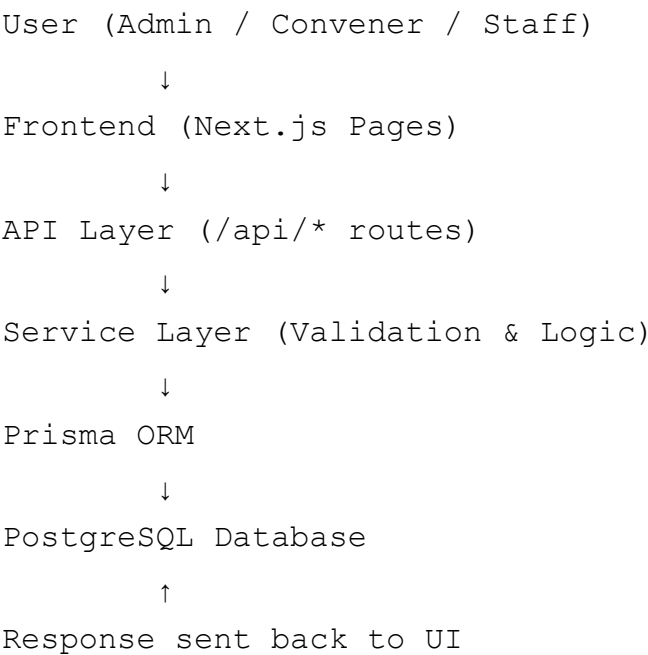
1. Actors Involved

- Admin
 - Meeting Convener
 - Staff
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2. System Components

- Frontend (Next.js UI)
 - Backend APIs (Next.js API Routes)
 - Service Layer (Business Logic)
 - Database (PostgreSQL using Prisma ORM)
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3. Overall Data Flow



4. Module-Wise Data Flow

4.1 User Authentication Flow

1. User enters login credentials on the UI
 2. Credentials are sent to `/api/auth/login`
 3. Backend validates user from the database
 4. Authentication response is returned
 5. User is redirected to the dashboard
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4.2 Meeting Creation Flow

1. Convener/Admin enters meeting details
 2. Data is sent to `/api/meetings`
 3. Service layer validates inputs
 4. Prisma stores meeting data in `meetings` table
 5. Confirmation response is sent to UI
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4.3 Attendance Management Flow

1. Convener selects meeting participants
 2. Data sent to `/api/meeting-members`
 3. Attendance stored in `meeting_member` table
 4. Updated attendance status returned to UI
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4.4 MOM Document Upload Flow

1. Convener uploads MOM document
 2. File is stored in external storage
 3. File path & metadata saved in `documents` table
 4. Success response sent to UI
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4.5 Report Generation Flow

1. User requests report from dashboard
2. Request sent to `/api/reports/*`

3. Backend fetches required data
 4. Report file generated (PDF/Excel)
 5. Report path stored in `reports` table
 6. Report delivered to user
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5. Security & Validation Flow

- Role-based access is checked at API level
- Input validation is handled in service layer
- Database access is controlled using Prisma ORM
- Only authorized users can access specific APIs